AGA KHAN UNIVERSITY EXAMINATION BOARD
HIGHER SECONDARY SCHOOL CERTIFICATE
CLASS XII EXAMINATION
APRIL/ MAY 2018

Computer Science Paper II

Time: 2 hours 25 minutes **Marks: 55**

Detailed Comments:

Constructed Response Questions (CRQs)

Question 1:

Define the following terms.

i. Database
ii. Table
iii. Query
iv. Field
v. Record

Question 2:

Fatima has created a database table named **Product**. Few records of this table are shown below.

Prod_ID	Name	Expire_Date	Imported	Price per Item
P101	Natural Milk	10/09/2018	☑	125
P102	Frozen Kebab	09/15/2018	☐	750
P103	Yogurt	01/26/2019	☑	140
P104	Frozen Fish	10/09/2018	☐	750
P105	Frozen Kebab	02/20/2019	☑	610

a. Write the data types she would have selected for the fields given below.

- Prod_ID
- Imported
- Price per Item

b. Write the name of the field from the given table which should be selected as a primary key. Give a valid reason to support your answer.

Question 3:

a. Describe ascending and descending sort.

b. Apply ascending sort on the given sequence of numbers. Show your working.
5^2, 1^9, 2^5, 4^2, 3^3

Question 4:

A database is set-up to store information about some of the high-rise buildings in Pakistan. Few records from the database are shown below.

Ref No.	Name of Building	City	Year	No. of Floors	Height (m)	Height (ft)
BIT100	Icon Tower	Karachi	2017	68	300	984
TO101	Riveria Mall	Lahore	2017	34	130	427
OT102	Sea Towers	Karachi	2012	30	120	394
CPT103	Centre Point Mall	Karachi	2013	28	117	384
TT104	Communication Tower	Islamabad	2011	25	113	371
TC105	The Centre Tower	Islamabad	2013	23	110	361
ASTP106	Information Technology Park	Lahore	2009	17	106	348

(**Note:** This is fictitious data about high-rise buildings in Pakistan.)

a. Show the output of the following query in design view of MS Access.

Field:	Ref No.	Name of Building	Year	Height (m)	Height (ft)
Table:					
Sort:					
Show:	☐	☑	☑	☑	☐
Criteria:			<=2013	>110	
or:					

Space for Output

b. Complete the query in design view below to select and show the Ref No., Name of Building, City, No. of Floors and Height (ft) of all buildings in Karachi or Lahore with height less than 450 feet.

Field:					
Table:					
Sort:					
Show:	☐	☐	☐	☐	☐
Criteria:					
or:					

Question 5a:

Describe ONE main characteristic for each type of given programming language which makes it different from the other two.

Machine Language	Assembly Language	

Question 5b:

What is the purpose of using **void** before **main( )** and **void** inside **brackets** in the given statement?

void main(void)

Using 'void' Before main( )	Using 'void' Inside Brackets

Question 6a:

What is the keyboard shortcut to run a C program in Turbo C integrated development environment (IDE)?

Question 6b:

Write a C program that would input values of radius (r) and height (h), calculate and output the surface area of the right circular cone.

The formula to calculate the surface area of the right circular cone is:

$$\text{Area} = \pi r\left(r + \sqrt{h^2 + r^2}\right) \text{ where } \pi = 3.142$$

Question 7:

a. Describe the purpose of using a loop in a program.

b. Write the general syntax of **while** loop and **if** statement.

c. What would be the output of the given code if the input is the first five integers, i.e. 1, 2, 3, 4, 5?

```
#include<stdio.h>
#include<conio.h>
void main( )
{
int n = 5;
float a, i, m, x = 0;
clrscr( );

for(i = 1; i <= n; i++)
{
 scanf("%f", &a);
 x = x + a;
}
m = x / n;
printf("\n%f", m);
getch( );
}
```

d. Infer the purpose of the given program.

Question 8:

Complete the given code for a user-defined function named **Square** which would calculate the square of any input number and invoke (call) it in the main function.
The formula to calculate the square of a number **n** is given below.
Square = n × n

```
#include<stdio.h>
//Declare square function below
                                        
int main( )
{
   float m, n;
   printf ( "\nEnter some number for finding square \n");
   scanf ( "%f", &m );
   // Call function below
                                        
 printf ( "\nSquare of the given number %f is %f",m,n );
}
 //Define function below
```

Question 9:

Read the given C language code.

```
#include <stdio.h>
#include <conio.h>
{
        FILE *fp;
        char ch;
        fp = fopen("MYFILE.TXT", "w");
        while((ch = getche( )) != '\r')
                putc(ch,fp);
        fclose(fp);
}
```

Interpret the purpose of the following code segments from this code.

i. FILE *fp;
ii. fp = fopen("MYFILE.TXT", "w");
iii. while((ch = getche( )) != '\r')
iv. putc(ch,fp);

Extended Response Questions (ERQs)

The following questions offered a choice between part **a** and **b**.

Question 10a:

Write a C program for calculating the roots of a quadratic equation using the quadratic formula. Also, determine whether the roots are real or imaginary. Apply suitable conditions and print an appropriate message.

The quadratic formula to calculate the roots of a quadratic equation is given below.

Roots = $\frac{-b \pm \sqrt{d}}{2a}$, where $d = b^2 - 4ac$; Roots are real when $d \geq 0$ otherwise roots are imaginary.

Question 10b:

Write a C program to accept the marks of a student for FIVE subjects, calculate and print the overall percentage and print the student's grade according to the criteria in the given table.

(**Note:** Total marks of subjects are 500.)

Percentage	Grade
80.0 to 100.0	A+
70.0 to 79.9	A
60.0 to 69.9	B
50.0 to 59.9	C
0 to 49.9	Fail